Anselm Long

+65 88536376 | anselmpius@gmail.com | linkedin.com/in/anselmlong | github.com/anselmlong | anselmlong.com

Education

National University of Singapore Singapore
Bachelor of Computing (Computer Science) Aug 2023 – May 2027

- **GPA:** 4.60 / 5.00. **Specialized Track:** Artificial Intelligence and Machine Learning
- **Relevant Coursework:** Artificial Intelligence and Machine Learning, Computer Vision, Software Engineering, Software Design and Architecture, Statistics and Probability, Computer Networks

Experience

Software Engineer Intern, Open Government Products (Maps) Aug 2026 – Feb 2027

- Shipped 4 production features across 29 merged PRs, including a Google Maps hand-off fix that raised place-search accuracy from 10% to 86% on 638 markers.
- Led a CSV-ingestion redesign and live map editing with AI column mapping powered by TypeSafe AI's **Jev** model, using **Next.js** and **TypeScript**, cutting estimated ingestion and mapping time by 75%.

Software Engineer Intern, Visa May 2026 – Aug 2026

- Developed SAGE, a RAG-powered incident investigation assistant using hybrid search (**BM25** + **pgvector**) and a Claude diagnosis agent, achieving 92% top-8 hit rate and 0.728 MRR on a 100-question benchmark.
- Engineered a data pipeline to consolidate 8,000+ email and company log records into 2,693 deduplicated incident clusters, and built a mainframe transaction trace analyzer to help operators understand transaction flows.
- Reduced incident investigation time by an estimated 40%, accelerating operator diagnosis and contributing to lower MTTR.

Lead Developer, Aegis Jan 2026 – Mar 2026

- Led 3 developers to build a full-stack, controlled AI-assistant platform for CTFs, supporting 80 concurrent users at the National Cybersecurity Olympiad with JWT auth, **RAG** chat, per-user token budgets and LLM guardrails.

Machine Learning Intern, IMDA May 2025 – Aug 2025

- Evaluated 8 anomaly-detection models on about 9M certificate records across 31 features, then tuned and ablated the top 6 with **scikit-learn** to reach 0.93 F1 on held-out data.

Projects

Optifiner | *Python, Multi-Agent, Orchestration* | optifiner.dev Jan 2026

- Won the Marshall Wace sponsor prize at Hack & Roll 2026 with an evolutionary **multi-agent** framework where parallel agents propose and optimize code changes and only the best survive.

Telegram Bot Fleet | *Python, TypeScript, Self-hosted* | bot.anselmlong.com 2025 – 2026

- Built and self-hosted Telegram bots with 900+ combined users, including hall aircon-credit alerts, daily mass readings, a 6-7 picture detector for group chats, and a personal assistant built on **Hermes**.

Freak-Cha | *YOLOv8, MediaPipe, Next.js* | freakcha.com Oct 2025

- Won Funniest Hack at HackHarvard 2025 (532 participants) with a CAPTCHA built in 36 hours, chaining a **YOLOv8** tongue detector with a **MediaPipe** face check.

Almost Anselm | *Mistral-7B, QLoRA, Axolotl* | anselmlong.com/blog/almost-anselm Nov 2025 – Dec 2025

- Fine-tuned a personal **Mistral-7B** LLM using **QLoRA** on 5,000 Telegram messages, cutting training time from an estimated 41 to about 3 hours and reaching a 28.2% human confusion rate in a blind test with 17 friends.

Skills

Languages: Python, TypeScript, JavaScript, Go, Java, C, C++
Frameworks: Next.js, React, FastAPI, Node.js, LangChain, Axolotl
AI/ML: PyTorch, scikit-learn, YOLOv8, RAG, LLM Fine-tuning
Databases & Tools: PostgreSQL, pgvector, Docker, Git, Playwright, Vercel

Interests and Activities

Interests: Photography, Videography, Bouldering, Coffee

Activities: NUS Climbing Club EXCO, Ridge View Residential College, Publicity and Social Media Team